Nigeria Energy Infrastructure & Electricity Access Analysis

- **OBJECTIVE**

This project was carried out to analyze electricity infrastructure distribution, power generation trends, and electricity access levels across Nigerian states with the use of SQL and Power BI. Its objective was to identify regional disparities in electricity infrastructure, identify dominant energy sources, and examine the relationship between infrastructure development and electricity accessibility

- **Tools Used**

❖ MySQL
❖ Power BI
❖ Microsoft Excel

- **Dataset Used and their Description**

❖ State electricity access dataset
❖ Grid infrastructure dataset
❖ Nigeria energy analysis dataset

❖ The electricity access data gives raw information on the electricity access rate (in percentage) of rural and urban areas in states across Nigeria, showing states with larger access than others for 5 years (2018 - 2023)

❖ The Infrastructure dataset gives raw information on the infrastructures allocated to each state, providing information like the infrastructure ID, state name, grid type, plant type, voltage capacity, both apparent and real power generated, transmission length, year commissioned, and functional status of these infrastructures across the states in Nigeria.

- The Nigeria energy analysis dataset is a merged analytical dataset containing information from both the electricity access and the infrastructure grid datasets. It merges both datasets based on their common column (state) to help generate insights and study trends for further analysis.

- **Data Cleaning Process**

Both datasets were first retrieved from an online source and imported into an Excel sheet, where I handled the null values before transferring them to a MySQL environment.
Here, both datasets were cleaned, standardized, aggregated, and summarised, and merged. View tables were also created for further analysis and then transferred to Power BI for visualization.

- **Key Insights**

The following insights were taken;

1. Lagos State dominates in the generation capacity sector in comparison to other states
2. Gas-powered plants dominate the energy mix
3. Higher infrastructure does not always guarantee higher electricity access
4. Electricity infrastructure varies significantly by state
5. Some states with relatively lower infrastructure still maintain moderate electricity access levels.

- **Dashboard Features**

These features were used to create the dashboard.

1. Interactive slicers and filters
2. Geographic map visualization
3. KPI performance cards
4. Infrastructure vs electricity access scatter plot analysis

5. Power generation ranking analysis
6. Energy mix analysis
7. Dynamic cross-visual interactions